SESSION 7

FINANCIAL PLANNING & ANALYSIS – 3

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

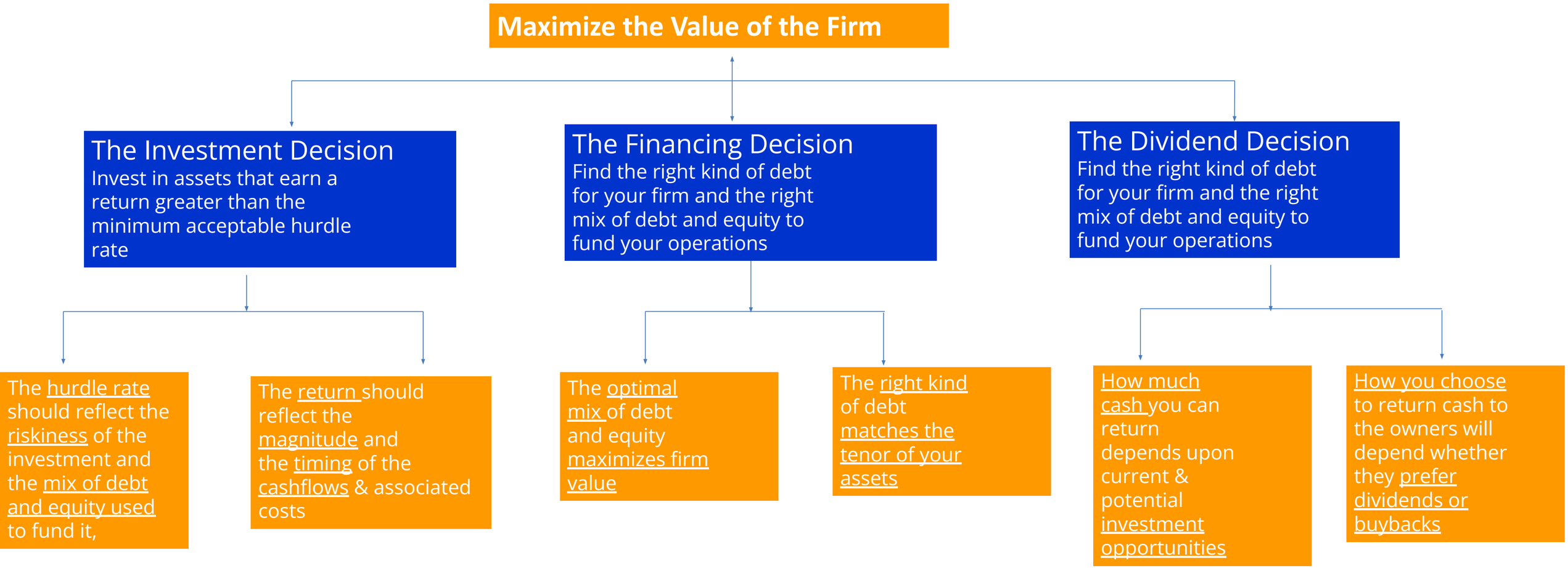
Session 7: Time Value of Money (TVM)

By the end of the session, learners will be able to:

- **Explain** the concept and importance of the Time Value of Money (TVM) in financial decision-making.
- **Explain** the concepts of Present Value (PV), Future Value (FV), compounding, and discounting.
- **Calculate** the Present Value and Future Value of single and multiple cash flows using appropriate TVM techniques.
- **Apply** TVM concepts to calculate the value of annuities and perpetuities.
- **Apply** appropriate discounting and compounding techniques to cash-flow streams and terminal values.
- Analyse different cash-flow patterns and determine their present or future values for basic engineering/business investment decisions.

Corporate Finance

Big Picture



Source: *Corporate Finance, Aswath Damodaran*

Time Value of Money (TVM)

Concepts

Situation 1:

I want to give you some money and ask that you choose from one of the following two alternatives:

- **₹10,000 today,**

OR

- **₹10,000 one year from now**

✔ **A mobile phone is available for ₹10,000 today.**
✔ **The price of chocolate remains at ₹10,000 after 1 year**

Question: So which option would you choose and why?

Time Value of Money (TVM)

Concepts

Situation 2:

I want to give you some money and ask that you choose one of the following two alternatives:

- **₹10,000 today,**

OR

- **₹11,000 one year from now**

✔ **A mobile phone is available for ₹10,000 today.**
✔ **The price of chocolate remains at ₹10,000 after 1 year**

Now, suppose you could earn 10% interest per year.

Question: Will your choice change? If so, why?

Time Value of Money (TVM)

Concepts

Situation 3:

I want to give you some money and ask that you choose one of the following **two alternatives:**

- **₹10,000 today,**

  OR

- **₹11,000 one year from now**

✔ **A mobile phone is available for ₹10,000 today.**
✔ **The price of the phone has increased to ₹12,000 after 1 year**

Now, the price of the phone has gone up by 20% in 1-years time.

The interest rate remains at 10% per annum.

Question: Will your choice change? If so, why?

Time Value of Money

Concepts

Factors affecting the Time Value of Money

- There is always a **trade-off** between **current consumption** & **future consumption**

  ✔ **Most of us prefer present consumption over future consumption**
- Investing today to generate returns and grow wealth over time.
- Over **time,** the **purchasing power of money** goes down due to inflation.
- There are **opportunity costs** associated with every decision.

Time Value of Money

Concepts

Future Value & Compounding

Suppose you invest ₹100 in a savings account that pays 10% interest per year. How much will you receive after 1 year?

- **You will have ₹110 = Original principal ₹100 + Interest ₹10**

Future Value — ₹110

Present Value — ₹100

If we invest for one time period at an interest rate 'r', the investment will grow to (1+r)

Or, **1 + 0.10 = 1.1** Rupees per Rupee invested

We invested **₹100** in our example, which will grow to **₹100 x 1.1 = ₹110**

Time Value of Money

Future Value & Compounding

Suppose you invest ₹100 in a savings account that pays 10% interest per year. How much will you receive in 2 years' & 3 years?

- **You will have ₹121 = Original principal ₹110 + Interest ₹10 + Interest on interest ₹1**

Future Value — **Present Value**

In 2 years', time

₹121 = ₹110 x 1.1
= (₹100 x 1.1) x 1.1
= ₹100 x (1.1 x 1.1)
= ₹100 x 1.1^2
= ₹100 x 1.21

In 3 years', time

₹133.10 = ₹121 x 1.1
= (₹110 x 1.1) x 1.1
= (₹100 x 1.1) x 1.1 x 1.1
= ₹100 x (1.1 x 1.1 x 1.1)
= ₹100 x 1.1^3
= ₹100 x 1.331

Future Value = \$1 x $(1+r)^t$

↓

Future value interest factor (FVIF)

Time Value of Money

A look at Simple Interest & Compound Interest

Using the same example of ₹100 invested @ 10% interest per year for 5 years. The Future Value of ₹100 under both Simple Interest and Compound Interest will be as follows:

Year	Beginning Amount (₹)	Simple Interest (₹)	Compound Interest ($)	Total Interest Earned (₹)	Ending Amount (₹)
1	100.00	10.00	0.00	10.00	110.00
2	110.00	10.00	1.00	11.00	121.00
3	121.00	10.00	2.10	12.10	133.10
4	133.10	10.00	3.31	13.31	146.41
5	146.41	10.00	4.64	14.64	161.05
		₹50	**₹11.05**	**₹61.05**	
		Total Simple Interest	**Total Compound Interest**	**Total Interest**	

Time Value of Money

The Power of Comppounding

Imagine if my great-grandfather had invested ₹100 in 1900 in my family name at 10%. How much would it be worth today?

Today, the investment would be worth,

$$\text{Future Value} = ₹1 \times (1+r)t$$

$$= ₹10 \times (1+r)^{t}$$

$$= ₹10 \times 1.1^{120}$$

$$= ₹10 \times 92709.07$$

$$= ₹927,090.7$$

Time Value of Money

Compounding vs. Discounting

COMPOUNDING

Moves value FORWARD in time — Present Value → Future Value



$$FV = PV \times (1 + r)^n$$

DISCOUNTING

Moves value BACKWARD in time — Future Value → Present Value



$$PV = FV \div (1 + r)^n$$

Where:

- PV = Present Value FV = Future Value
- r = interest / discount rate per period n = number of periods
- $(1 + r)^n$ is the compounding factor; $1 \div (1 + r)^n$ is the discounting (PV) factor

Time Value of Money

Key Notations:

Symbol	Meaning
PV	Present Value — value of a cash flow today
FV	Future Value — value of a cash flow at a future date
CF	Cash Flow — an amount received or paid at a point in time
r	Interest rate / discount rate / required rate of return per period
n	Number of compounding / discounting periods
g	Constant growth rate of cash flows (used in growing annuity/perpetuity)
A	Annuity payment — equal periodic cash flow

Time Value of Money

Future Value of a Single Cash Flow

$$FV = PV \times (1 + r)^n$$

Compounding formula



WORKED EXAMPLE

A student deposits ₹10,000 in a savings scheme earning 10% p.a. compounded annually. What will it be worth after 3 years?

- PV = ₹10,000, r = 10%, n = 3
- FV = 10,000 × $(1.10)^3$
- FV = 10,000 × 1.331

FV = ₹13,310

The deposit grows by ₹3,310 purely through compounding.

Time Value of Money

Annual Percentage Rate (APR)

Question: A credit card company charges a nominal rate of 2% per month. If a customer has purchased ₹100 worth of goods on her credit card, calculate the amount she will owe after 1 year and the annual percentage rate (APR).

The amount (FV) she will owe at the end of the nth year is given by:

$$FV = PV \times (1+r)^n$$

Where PV= Principal or present value; r=rate of interest; n= number of periods or years

Here PV= ₹100; r=2% or 0.02; n= 12

$$FV = 100 \times (1+0.02)^{12}$$

$$FV = 100 \times 1.2682$$

$$FV = ₹126.82$$

APR = 24% (2 x 12) ✖

Annual percentage rate (APR) : {(₹126.82 - ₹100)/ ₹100}*100

APR = 26.82%

Time Value of Money

Discounting

- What will the ₹10,000 investment grow to if it earns a 7.5% return every year for the next 5 years? **(Verify that the answer is about ₹14356).**

Another type of question

- I will need ₹10,000 in 5 years, and the interest rate is 7.5% now. How much do I need to invest now?

In other words, we know that the future value here is ₹10000.
We want to know what the present value is, or the value of the money today.

$$FV = PV(1+r)^n$$

$$PV = FV/(1+r)^n$$

Where **PV= Principal or present value; r=rate of interest; n= number of periods or years**

Time Value of Money

Discounting

Another type of question

- I will need ₹10,000 in 5 years, and the interest rate is 7.5% now. How much do I need to invest now?

In other words, we know that the future value here is ₹10,000

We want to know what the present value, or value of the money today?

$FV = PV(1+r)^n$

$PV = FV/(1+r)^n$

$PV = ₹10000/(1+0.075)^5$

$PV = ₹10000/(1.075)^5$

$PV = ₹6965.58$

$1/(1+r)^n$ is known as the discounting factor

Time Value of Money

Discounting

- What amount now is equivalent to ₹9000 in 5 years, with interest at 16.50% p.a.?

In other words, we know that the future value here is ₹9000.
We want to know what the present value, or the value of the money today?

Using Algebra

$A \times (1+0.165)^5 = 9000$

$A \times (1.165)^5 = 9000$

$A = ₹9000/(1.165)^5$

$A = ₹4193.85$

$FV = PV(1+r)^n$

$PV = FV/(1+r)^n$

$PV = ₹9000/(1+0.165)^5$

$PV = ₹9000/(1.165)^5$

$PV = ₹4193.85$

$1/(1+r)^n$ is known as the discounting factor

Time Value of Money

Discounting

- What is the present value of ₹1,200 receivable in 12 years, with interest at 13%?

In other words, we know that the future value here is ₹1200.
We want to know what the present value, or the value of the money today?

Using Algebra

$A \times (1+0.13)^{12} = 1200$

$A \times (1.13)^{12} = 1200$

$A = ₹1200/(1.13)^{13}$

$A = ₹276.85$

$FV = PV(1+r)^n$

$PV = FV/(1+r)^n$

$PV = ₹1200/(1+0.13)^{12}$

$PV = ₹1200/(1.13)^{12}$

$PV = ₹276.85$

$1/(1+r)^n$ is known as the discounting factor

Time Value of Money

Discounting & Compounding



Future value factor = $(1+r)^n$

Present value factor = $1/(1+r)^n$

Time Value of Money

Present Value of a Single Cash Flow

$$PV = FV \div (1 + r)^n$$

Discounting formula



WORKED EXAMPLE

You are promised ₹50,000 exactly 5 years from now. If your required rate of return is 8% p.a., what is this worth today?

- FV = ₹50,000, r = 8%, n = 5
- PV = 50,000 ÷ $(1.08)^5$
- PV = 50,000 ÷ 1.4693

PV = ₹34,029

₹34,029 today is equivalent to ₹50,000 five years later at 8%.

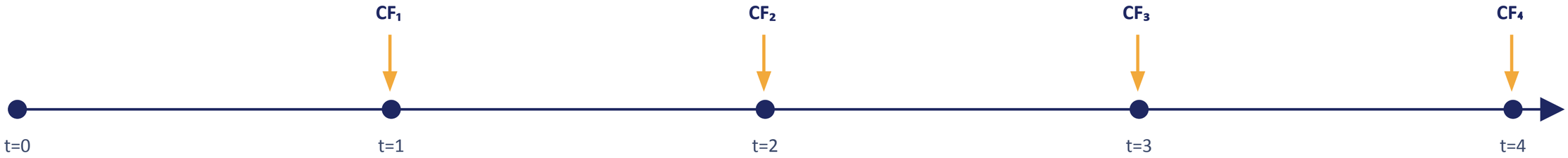
Time Value of Money

Valuing Multiple (Uneven) Cash Flows

Real projects rarely generate a single payment. When cash flows differ each period, value each one separately and sum the results.

$$PV = \Sigma \ CF\square \div (1 + r)^t$$

Sum of the present value of every individual cash flow



- Each cash flow is discounted using its own time period t
- Common in equipment cash-flow schedules, project appraisals and loan repayments with irregular instalments

Time Value of Money

Example: Multiple Cash Flows

A machine is expected to generate the following year-end cash flows. Discount rate = 10%. Find the total present value.

Year (t)	Cash Flow (₹)	PV Factor @10% $1/(1.10)^t$	Present Value (₹)
1	5,000	0.9091	4,545.45
2	6,000	0.8264	4,958.68
3	7,000	0.7513	5,259.95
4	8,000	0.6830	5,464.23

Total Present Value = ₹4,545.45 + ₹4,958.68 + ₹5,259.95 + ₹5,464.23 = ₹20,228.31

This ₹20,228.31 today is equivalent to receiving the full uneven stream of cash flows.

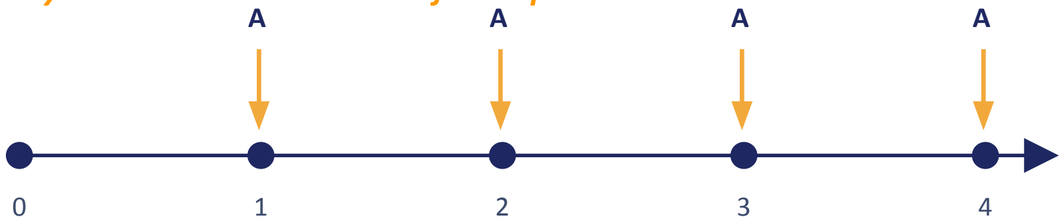
Time Value of Money

Annuity: Equal Periodic Cash Flows

An annuity is a series of equal cash flows occurring at regular intervals for a fixed number of periods.

ORDINARY ANNUITY

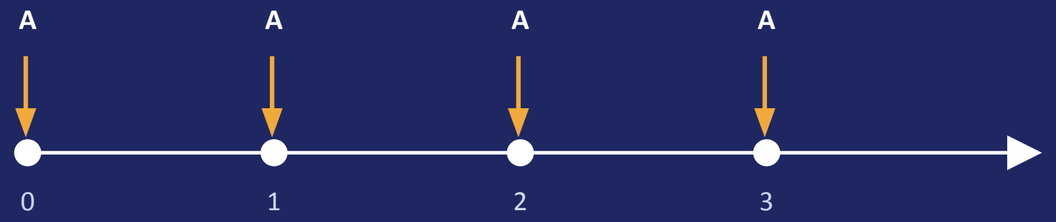
Payments occur at the END of each period



- ***Example: Loan EMIs, most bond coupon schedules***

ANNUITY DUE

Payments occur at the START of each period



- Example: Rent, insurance premiums paid in advance

Time Value of Money

Annuity: Present & Future Value Formulas

$$PV = A \times [1 - (1+r)^{-n}] \div r$$

Present value of an ordinary annuity

$$FV = A \times [(1+r)^{n} - 1] \div r$$

Future value of an ordinary annuity

Annuity Due = Ordinary Annuity value × (1 + r)

- A = the equal periodic annuity payment; r = periodic rate; n = number of payments
- The term $[1 - (1+r)^{-n}] \div r$ is called the **Present Value Interest Factor of Annuity (PVIFA)**
- The term $[(1+r)^{n} - 1] \div r$ is called the **Future Value Interest Factor of Annuity (FVIFA)**
- Because an annuity-due payment arrives one period earlier, its value is simply the ordinary-annuity value compounded forward by one period

Time Value of Money

Example: Ordinary Annuity

An engineering graduate plans to invest ₹10,000 at the end of every year for 5 years in a fund earning 10% p.a.

PRESENT VALUE OF THE ANNUITY

- A = ₹10,000, r = 10%, n = 5
- PV = $10{,}000 \times [1 - (1.10)^{-5}] \div 0.10$
- **PV = 10,000 × 3.7908**

PV = ₹37,908

Lump sum today equivalent to the 5 annual deposits

FUTURE VALUE OF THE ANNUITY

- A = ₹10,000, r = 10%, n = 5
- FV = $10{,}000 \times [(1.10)^{5} - 1] \div 0.10$
- **FV = 10,000 × 6.1051**

FV = ₹61,051

Maturity value of the fund at the end of year 5

Time Value of Money

Perpetuity: Cash Flows That Never End (Forever)

A perpetuity is an annuity that continues forever with no maturity date.

$$PV = CF \div r$$

Constant (level) perpetuity

$$PV = CF_1 \div (r - g)$$

Growing perpetuity (Gordon model), r > g

EXAMPLE: CONSTANT PERPETUITY

- A preference share pays ₹500 every year forever
- Required return r = 10%
- PV = 500 ÷ 0.10

$$PV = ₹5{,}000$$

EXAMPLE — GROWING PERPETUITY

- First payment ₹500, growing at g = 4% p.a. forever
- Required return r = 10%
- PV = 500 ÷ (0.10 − 0.04)

$$PV = ₹8{,}333$$

Time Value of Money

Terminal Value: Valuing Cash Flows Beyond the Forecast

In project or business valuation, cash flows are forecast explicitly for a few years; **Terminal Value (TV)** captures the value of **ALL cash flows beyond that horizon,** assuming they grow at a stable rate forever.

$$TV_n = [CF_n \times (1 + g)] \div (r - g)$$

Terminal value at the end of the explicit forecast period n (Gordon Growth Model)

$$\text{Present Value of Terminal Value} = TV_n \div (1 + r)^n$$

- **CF_n =** the last explicitly forecast cash flow (year n); **g = long-term stable growth rate; r = discount rate**
- **Terminal value** must itself be **discounted back to today** since it is valued as of year n
- Widely used in capital budgeting and DCF valuation of projects, plants and equipment, and businesses

Time Value of Money

Example: Terminal Value

An investment project's explicit forecast ends in Year 5 with a cash flow of ₹2,00,000. Beyond Year 5, cash flows are expected to grow at 3% p.a. forever. Discount rate = 12%.

STEP 1 — TERMINAL VALUE AT YEAR 5

- $TV_5 = [2,00,000 \times 1.03] \div (0.12 - 0.03)$
- $TV_5 = 2,06,000 \div 0.09$

TV_5 = ₹22,88,889

STEP 2 — DISCOUNT TV TO TODAY

- PV of TV = $22,88,889 \div (1.12)^5$
- PV of TV = 22,88,889 ÷ 1.7623

PV of Terminal Value = ₹12,98,590

Time Value of Money

Summary

Cash-Flow Type	Formula	Application
Single Cash Flow	$FV = PV(1+r)^n$ / $PV = FV \div (1+r)^n$	Lump-sum equipment purchase / resale value
Multiple Cash Flows	$PV = \Sigma CF_t \div (1+r)^t$	Uneven project cash-flow schedules
Annuity	$PV = A[1-(1+r)^{-n}] \div r$	Loan EMIs, AMC & lease payments
Perpetuity	$PV = CF \div r$ (or $CF \div (r-g)$)	Perpetual maintenance / endowment funds
Terminal Value	$TV = CF_n(1+g) \div (r-g)$	Long-life project / plant valuation beyond forecast

Thank You!